Economics Letters 216 (2022) 110604

Contents lists available at ScienceDirect

Economics Letters

journal homepage: www.elsevier.com/locate/ecolet

Interpreting the coefficients in dynamic two-way fixed effects regressions with time-varying covariates

Lihua Lin, Zhengyu Zhang *

School of Economics, Shanghai University of Finance and Economics, China

ARTICLE INFO

Article history:
Received 19 March 2022
Received in revised form 28 April 2022
Accepted 9 May 2022
Available online 16 May 2022

JEL classification:
C13
C23
C31

Keywords:
Two-way fixed effects regression
Heterogeneous treatment effects
Time varying covariates
Conditional parallel trends assumption

ABSTRACT

This paper discusses the causal interpretation of event study coefficients in a dynamic two-way fixed effects regression with time-varying covariates under a conditional parallel trends assumption. In addition to the "cross-lag contamination" problem noted by Sun and Abraham (2021), we find a new source of bias induced by the time-varying effect of the covariates. This covariate effect bias will not be eliminated even under treatment effect homogeneity and will remain in the canonical two-period difference-in-differences setting unless one makes more restrictive assumptions on the effects of covariates.

1. Introduction

Researchers are often interested in dynamic treatment effects in a two-way fixed effects (TWFE) specification that resembles the following:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell} \mu_\ell \mathbf{1}\{t - E_i = \ell\} + v_{i,t}, \tag{1.1}$$

where $Y_{i,t}$ is the outcome of interest for unit i at time t; α_i and λ_t are unit and time fixed effects, respectively; and E_i is the time when unit i initially receives the binary absorbing treatment. $\ell = t - E_i$ denotes the relative period since treatment. Practitioners typically use estimate μ_ℓ as a reasonable measure of the dynamic treatment effect.

However, recent advances show that μ_ℓ may not be causally interpreted in settings with variations in treatment timing and heterogeneous treatment effects (e.g., de Chaisemartin and D'Haultfoeuille (2020), Borusyak et al. (2021), Goodman-Bacon (2021), Sun and Abraham (2021), Athey and Imbens (2022). In particular, Sun and Abraham (2021), hereinafter "SA") decompose μ_ℓ as a linear combination of cohort-specific effects from both its own relative period ℓ and other relative periods; unless strong assumptions regarding treatment effect homogeneity hold, effects from other relative periods will contaminate estimate μ_ℓ. We refer to this as "cross-lag contamination". This does not exist in the canonical two-period DID.

SA investigate the causal interpretation of μ_ℓ in (1.1) by imposing an unconditional version of the parallel trends assumption (PTA). In practice, applied researchers often incorporate time-varying covariates to control for confounders that vary across time and units.

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\ell} \mu_\ell \mathbf{1}\{t - E_i = \ell\} + \beta^\intercal \mathbf{X}_{i,t} + v_{i,t}, \tag{1.2}$$

where $\mathbf{X}_{i,t}$ is a vector of the covariates of unit i in period t. Our paper extends the existing literature in two ways. First, we decompose μ_ℓ in (1.2) by explicitly accounting for the presence of time-varying covariates under a conditional parallel trend assumption. Our PTA requires that the baseline outcomes would follow parallel paths for each cohort, conditional on covariates in all time periods. We find that in addition to cross-lag contamination, a new source of bias appears induced by the inclusion of covariates. This covariate effect bias will not vanish even under treatment effect homogeneity and will remain in a simple two-period setting. Moreover, we show that the covariate effect bias is eliminated as long as the effects of covariates are constant over time.

The remainder of this paper proceeds as follows. Section 2 introduces the event study design and provides the main results regarding the population regression coefficients of the dynamic TWFE regression with time-varying covariates. Section 3 discusses the sufficient assumptions for these coefficients to be

* Correspondence to: No. 111, Wuchuan Road, 200433, Shanghai, China.
E-mail address: zy.zhang@mail.shufe.edu.cn (Z. Zhang).

https://doi.org/10.1016/j.econlet.2022.110604

causally interpretable. Section 4 concludes the paper. Proofs as well as a small scale Monte Carlo simulation are presented in the supplementary material.

2. Main results

Similar to the setting used in SA, we consider a sample of N units observed over $T+1$ periods, where T is fixed. For each unit $i \in \{1, \ldots, N\}$ and period $t \in \{0, \ldots, T\}$, we observe the outcome variable $Y_{i,t}$, a vector of covariates $\mathbf{X}_{i,t}$, and the treatment status $D_{i,t} \in \{0, 1\}$: $D_{i,t} = 1$ if i is treated in period t, and 0 otherwise. We focus on the absorbing (staggered) treatment design, where $D_{i,t-1} = 1$ implies that $D_{i,t} = 1$ for all $t = 1, \ldots, T$. We denote the initial treatment period for unit i by $E_i := \min\{t, D_{i,t} = 1\}$. If unit i is never treated, we set $E_i = \infty$. We can uniquely categorize the units into disjoint cohorts e for $e \in \{0, \ldots, T, \infty\}$ based on the initial treatment period.

Let $Y_{i,t}^{e}$ be the potential outcome in period t, when unit i is first treated in time period e. Let $Y_{i,t}^{\infty}$ be the potential outcome if unit i never receives the treatment, which is called the baseline outcome. We can represent the observed outcome as:

$$Y_{i,t} = Y_{i,t}^{E_i} = Y_{i,t}^{\infty} + \sum_{e=0}^{T} \left(Y_{i,t}^{e} - Y_{i,t}^{\infty}\right) 1\{E_i = e\}.$$

Unlike SA, we consider the specification with covariates

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \mu_g \mathbf{1}\{t - E_i \in g\} + \beta^{\intercal} \mathbf{X}_{i,t} + v_{i,t} \tag{2.1}$$

where set $\mathcal{G}$ collects disjoint sets g of relative periods $\ell \in [-T, T]$. If all g's are singletons of relative time periods, then (2.1) becomes the fully dynamic specification. We allow some relative periods to be excluded from the specification and denote the excluded set with $g^{excl} = \{\ell : \ell \notin \bigcup_{g \in \mathcal{G}} g\}$.

We introduce some notation. Let $\ddot{Z}_{i,t}$ be time and cross-sectional demeaned version of $Z_{i,t}$, defined as $\ddot{Z}_{i,t} = Z_{i,t} - E[Z_{i,t}] - \frac{1}{T+1}\sum_{s=0}^{T} Z_{i,s} + \frac{1}{T+1}\sum_{s=0}^{T} E[Z_{i,s}]$ where the expectation $E[Z_{i,t}]$ is taken cross-sectionally at time $t = 0, \ldots, T$. Thus, $\ddot{Z}_{i,t}$ is referred as the double-demeaned version of $Z_{i,t}$. Let $\xi_{i,t}^{g}$ be the population residual from regressing $\mathbf{1}\{t - E_i \in g\}$ on the covariates $\mathbf{X}_{i,t}$ and the two-way fixed effects. Collect $\ddot{\xi}_{i,t}^{g}$ and the relative time indicators in (2.1) in column vectors $\xi_{i,t} = (\xi_{i,t}^{g})_{g \in \mathcal{G}}^{\intercal}$ and $\mathbf{D}_{i,t} = (1\{t - E_i \in g\})_{g \in \mathcal{G}}^{\intercal}$, respectively.

Assumption 1.

(i) (Sampling shceme) The observations $\{Y_{i,0}, \ldots, Y_{i,T}, \mathbf{X}_{i,0}, \ldots, \mathbf{X}_{i,T}, D_{i,0}, \ldots, D_{i,T}\}$ are independent and identically distributed.

(ii) (Existence of moments of variables) $E\left[\ddot{\mathbf{X}}_{i,t}\ddot{\mathbf{X}}_{i,t}^{\intercal}\right] < \infty$, $E\left[\ddot{\mathbf{X}}_{i,t}\ddot{\mathbf{D}}_{i,t}^{\intercal}\right] < \infty$, $E\left[\ddot{\mathbf{D}}_{i,t}\ddot{\mathbf{D}}_{i,t}^{\intercal}\right] < \infty$, $E\left[\ddot{\xi}_{i,t}\ddot{\mathbf{D}}_{i,t}^{\intercal}\right] < \infty$, $E\left[\ddot{\xi}_{i,t}\ddot{\xi}_{i,t}^{\intercal}\right] < \infty$ and $E\left[\ddot{\xi}_{i,t}\ddot{Y}_{i,t}\right] < \infty$.

(iii) (Rank conditions) $\sum_{t=0}^{T} E\left[\ddot{\mathbf{X}}_{i,t}\ddot{\mathbf{X}}_{i,t}^{\intercal}\right]$, $\sum_{t=0}^{T} E\left[\ddot{\xi}_{i,t}\ddot{\xi}_{i,t}^{\intercal}\right]$ and $\begin{pmatrix} \sum_{t=0}^{T} E\left[\ddot{\mathbf{D}}_{i,t}\ddot{\mathbf{D}}_{i,t}^{\intercal}\right] & \sum_{t=0}^{T} E\left[\ddot{\mathbf{D}}_{i,t}\ddot{\mathbf{X}}_{i,t}^{\intercal}\right] \\ \sum_{t=0}^{T} E\left[\ddot{\mathbf{X}}_{i,t}\ddot{\mathbf{D}}_{i,t}^{\intercal}\right] & \sum_{t=0}^{T} E\left[\ddot{\mathbf{X}}_{i,t}\ddot{\mathbf{X}}_{i,t}^{\intercal}\right] \end{pmatrix}$ have full rank.

The sampling shceme is the same as Sant'Anna and Zhao (2020), Callaway and Sant'Anna (2021), SA, and Athey and Imbens (2022). It imposes that each unit i is randomly drawn from a large population and does not restrict temporal dependence of the observations. Finiteness of $E\left[\ddot{\xi}_{i,t}\ddot{\mathbf{D}}_{i,t}^{\intercal}\right]$ and $E\left[\ddot{\xi}_{i,t}\ddot{\xi}_{i,t}^{\intercal}\right]$ is not necessary because they can be derived from the remainder of Assumption 1. We keep them simply for convenience. Rank conditions make the corresponding matrices invertible. SA maintains a similar assumption without covariates.

First, we provide a basic decomposition result for μ_g, which only relies on Assumption 1. Let $\mathbf{X}_i = (\mathbf{X}_{i,0}, \ldots, \mathbf{X}_{i,T})$.

Proposition 2.1. *Under Assumption 1, the population two-way fixed effects regression coefficient μ_g in (2.1) can be written as*

$$\begin{aligned}
\mu_g = & \sum_{\ell \in g}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g} \cdot \left(E\left[Y_{i,e+\ell} - Y_{i,0}^{\infty} | \mathbf{X}_i, E_i = e\right]\right.\right. \\
& \left.\left. - E\left[Y_{i,e+\ell}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]\right)\right] \\
& + \sum_{g' \neq g, g' \in \mathcal{G}}\sum_{\ell \in g'}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g} \cdot \left(E\left[Y_{i,e+\ell} - Y_{i,0}^{\infty} | \mathbf{X}_i, E_i = e\right]\right.\right. \\
& \left.\left. - E\left[Y_{i,e+\ell}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]\right)\right] \\
& + \sum_{\ell \in g^{excl}}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g} \cdot \left(E\left[Y_{i,e+\ell} - Y_{i,0}^{\infty} | \mathbf{X}_i, E_i = e\right]\right.\right. \\
& \left.\left. - E\left[Y_{i,e+\ell}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]\right)\right] \\
& + \sum_{t=0}^{T} E\left[\omega_{i,\infty,t}^{g} \cdot \left(E\left[Y_{i,t} - Y_{i,0}^{\infty} | \mathbf{X}_i, E_i = \infty\right]\right.\right. \\
& \left.\left. - E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]\right)\right] \\
& + \sum_{t=0}^{T} E\left[\omega_{i,t}^{g} \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]\right]
\end{aligned}$$

where $\omega_{i,t}^{g} := \mathbf{e}_g^{\intercal}\left(\sum_{t=0}^{T} E\left[\ddot{\xi}_{i,t}\ddot{\xi}_{i,t}^{\intercal}\right]\right)^{-1}\ddot{\xi}_{i,t}$. The exact expressions of $\omega_{i,e,\ell}^{g}$, $e = 0, \ldots, T, \infty$ are deferred to the Appendix. Moreover,

$$\sum_{\ell \in g}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] = 1, \sum_{\ell \in g'}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] = 0 (\forall g' \neq g, g' \in \mathcal{G}),$$

$$\sum_{\ell \in g^{excl}}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] = -1$$

$$E\left[\omega_{i,t}^{g}\right] = \sum_{t=0}^{T} \omega_{i,t}^{g} = 0.$$

The aforementioned proposition shows that μ_g is a linear combination of five parts: (i) differences in the trends from its own relative period $\ell \in g$; (ii) differences in trends from relative periods $\ell \in g'$ belonging to other bins $g' \neq g$ but are included in the specification; (iii) differences in trends from relative periods excluded in specification $\ell \in g^{excl}$; (iv) differences in trends for never-treated units; and (v) trends of baseline outcomes conditional on covariates. While the first four parts have already been suggested by SA (proposition 1, page 181), the last one, $\sum_{t=0}^{T} E\left[\omega_{i,t}^{g} \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]\right]$, can be viewed as a new source of bias owing to the presence of covariates. In general, $E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty} | \mathbf{X}_i\right]$ varies across both i and t. If $\mathbf{X}_i = 1$, that is, there is no additional covariate to control for, the last part becomes

$$\sum_{t=0}^{T} E\left[\omega_{i,t}^{g} \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty}\right]\right] = \sum_{t=0}^{T} E\left[\omega_{i,t}^{g}\right] \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty}\right] = 0.$$

Assumption 2 (*Parallel Trends Conditional on Strictly Exogenous Covariates*)**.** For all $t \in \{1 \cdots, T\}$, $E\left[Y_{i,t}^{\infty} - Y_{i,t-1}^{\infty} | \mathbf{X}_i, E_i = e\right]$ is the same for all $e \in supp(E_i)$.

Assumption 2 requires that the baseline outcomes would follow parallel paths for each cohort, conditional on covariates *in all time periods*. Compared with the unconditional PTA in SA, our Assumption 2 is more plausible if the observed covariates that are thought to be correlated with the dynamics of the baseline outcome are unbalanced among cohorts. The motivation to condition on a set of covariates is similar to the PTAs conditional on pre-treatment covariates in Abadie (2005), Sant'Anna and Zhao (2020), and Callaway and Sant'Anna (2021).

One may be concerned that including covariates measured after treatment may obscure the causal interpretation of μ_g. However, when one estimates (2.1), they implicitly assume that the temporal change of $\mathbf{X}_{i,t}$ is not caused by the policy, which we call "strict exogeneity of $\mathbf{X}_i$". Otherwise, an over-control bias may exist by controlling the mediator variable on the indirect causal path $D_{i,t} \to \mathbf{X}_{i,t} \to Y_{i,t}$.

Consider the following linearly generated baseline outcomes: $Y_{i,t}^{\infty} = \alpha_i + \lambda_t + \beta_t^{\intercal}\mathbf{X}_{i,t} + v_{i,t}$. Assumption 2 implies that $E\left[v_{i,t} - v_{i,t-1}|\mathbf{X}_i, E_i = e\right]$ is the same across cohorts. This is plausible when $v_{i,t}$ follows a random walk.

In addition, de Chaisemartin and D'Haultfoeuille (2020) propose a similar assumption (online appendix, assumption S4, page 8) to obtain decomposition results for regressions with time-varying covariates. Under no anticipatory behavior, it can be written as

$$E\left[Y_{i,t}^{\infty} - Y_{i,t-1}^{\infty}|\mathbf{X}_i, E_i = e\right] = \gamma^{\intercal}\left(\mathbf{X}_{i,t} - \mathbf{X}_{i,t-1}\right) + \lambda_t$$

Our Assumption 2 is weaker because it allows different coefficients before $\mathbf{X}_{i,t}$ and $\mathbf{X}_{i,t-1}$; namely,

$$E\left[Y_{i,t}^{\infty} - Y_{i,t-1}^{\infty}|\mathbf{X}_i, E_i = e\right] = \gamma_t^{\intercal}\mathbf{X}_{i,t} - \gamma_{t-1}^{\intercal}\mathbf{X}_{i,t-1} + \lambda_t$$

Similar to SA, we define $CATT_{e,\ell}(\mathbf{X}_i) = E\left[Y_{i,e+\ell} - Y_{i,e+\ell}^{\infty}|E_i = e, \mathbf{X}_i\right]$ as the average treatment effect ℓ periods from the initial treatment for the cohort of units first treated at time e conditional on covariates in all time periods. Let $e_0 \in supp(E_i)$.

Proposition 2.2. *Under Assumptions 1–2, we can decompose μ_g as*

$$\begin{aligned}\mu_g =& \sum_{\ell\in g}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g} \cdot CATT_{e+\ell}(\mathbf{X}_i)\right] \\ &+ \sum_{g'\neq g, g'\in\mathcal{G}}\sum_{\ell\in g'}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g} \cdot CATT_{e+\ell}(\mathbf{X}_i)\right] \\ &+ \sum_{\ell\in g^{excl}}\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g} \cdot CATT_{e+\ell}(\mathbf{X}_i)\right] \\ &+ \sum_{t=0}^{T} E\left[\omega_{i,t}^{g} \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty}|\mathbf{X}_i, E_i = e_0\right]\right]\end{aligned}$$

Under Assumptions 1–2, the terms in Proposition 2.1 reduce to a linear combination of the causally interpretable building block $CATT_{e,\ell}$ and trends of baseline outcomes conditional on covariates for cohort e_0. We define $ATT_\ell = \sum_{e\in supp(E_i)} Pr\{E_i = e\} \cdot E\left[Y_{i,e+\ell} - Y_{i,e+\ell}^{\infty}|E_i = e\right]$ as the weighted average of treatment effect from ℓ periods relative to the treatment, with weights corresponding to cohort shares.

Assumption 3 (*Treatment Effect Homogeneity*)**.** For each relative period ℓ, $CATT_{e,\ell}(\mathbf{X}_i)$ does not depend on cohort e and covariates $\mathbf{X}_i$. That is $CATT_{e,\ell}(\mathbf{X}_i) = ATT_\ell$.

Assumption 3 requires homogeneity similar to that of SA (assumption 3, page 178). Moreover, it does not preclude dynamic treatment effects.

Proposition 2.3. *Under Assumptions 1–3, we can decompose μ_g as*

$$\begin{aligned}\mu_g =& \sum_{\ell\in g}\left(\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] ATT_\ell\right) \\ &+ \sum_{g'\neq g, g'\in\mathcal{G}}\sum_{\ell\in g'}\left(\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] ATT_\ell\right) \\ &+ \sum_{\ell\in g^{excl}}\left(\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] ATT_\ell\right) \\ &+ \sum_{t=0}^{T} E\left[\omega_{i,t}^{g} \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty}|\mathbf{X}_i, E_i = e_0\right]\right]\end{aligned}$$

Comparing Proposition 2.3 with the decomposition in SA (proposition 4, page 183), the last term, which is related to trends of baseline outcomes conditional on covariates for cohort e_0, complicates the causal interpretation of μ_g under treatment effect homogeneity.

Remark 2.1. To represent decomposition concisely, we consider the fully dynamic specification

$$\begin{aligned}Y_{i,t} =& \alpha_i + \lambda_t + \sum_{\ell=-K}^{-2} \mu_\ell \mathbf{1}\{t - E_i = \ell\} \\ &+ \sum_{\ell=0}^{L} \mu_\ell \mathbf{1}\{t - E_i = \ell\} + \beta^{\intercal}\mathbf{X}_{i,t} + v_{i,t}\end{aligned} \tag{2.2}$$

If we assume zero treatment effects for distant relative periods, i.e., $ATT_\ell = 0, \ell \in \{\cdots, -K-2, -K-1, L, L+1\cdots\}$, and no anticipatory behavior, i.e., $ATT_{-1} = 0$, then $ATT_{\ell'} = 0, \ell' \in g^{excl}$.

Under Assumptions 1–3 combined with $ATT_{\ell'} = 0, \ell' \in g^{excl}$, the decomposition simplifies to

$$\mu_\ell = ATT_\ell + \sum_{t=0}^{T} E\left[\omega_{i,t}^{\ell} \cdot E\left[Y_{i,t}^{\infty} - Y_{i,0}^{\infty}|\mathbf{X}_i, E_i = e_0\right]\right]$$

Thus, μ_ℓ cannot be causally interpreted in a fully dynamic specification even under treatment effect homogeneity and zero treatment effects for excluded periods.

3. When the covariate effect bias can be eliminated

The preceding analysis implies that, in general, the estimand of coefficients in a TWFE regression contains a covariate-induced bias. This section discusses the circumstances under which such bias can be eliminated.

Assumption 4 (*Time-Invariant Effect of Covariates on Baseline Outcomes*)**.** $E\left[Y_{i,t}^{\infty}|\mathbf{X}_i, E_i = e_0\right] = E\left[Y_{i,t}^{\infty}|\mathbf{X}_{i,t}, E_i = e_0\right] = a_t + b^{\intercal}\mathbf{X}_{i,t}$ for $t = 0, 1, \ldots, T$

Assumption 4 has two implications. First, among the covariates in all time periods, the baseline outcome for cohort $E_i = e_0$ in a particular period is only linearly correlated to the contemporary covariates. This linearity is not as strong as it appears because one may include high-order and interactive terms in $\mathbf{X}_{i,t}$. In addition, it assumes a time-invariant effect of covariates.

Consistent with Assumption 4's intuition to restrict the temporal effect of covariates, Assumption 5 is also proposed to eliminate the covariate effect bias.

Assumption 5. $E\left[Y_{i,t}^{\infty} - Y_{i,t-1}^{\infty}|\mathbf{X}_i, E_i = e_0\right] = (a_t - a_{t-1}) + b^{\intercal}\left(\mathbf{X}_{i,t} - \mathbf{X}_{i,t-1}\right)$ for $t = 1, \ldots, T$.

Assumption 5 requires the same effect of $\mathbf{X}_{i,t}$ and $\mathbf{X}_{i,t-1}$ on $Y_{i,t}^{\infty} - Y_{i,t-1}^{\infty}$. This is weaker than Assumption 4 because it restricts the difference in baseline outcomes, which cancels out the individual-specific time-invariant characteristics.

Specifically, consider the DGP for cohort $E_i = e_0$, $Y_{i,t}^{\infty} = \alpha_i + \lambda_t + \beta_t^{\intercal}\mathbf{X}_{i,t} + v_{i,t}$. Assumption 5 implies that $E\left[v_{i,t}|\mathbf{X}_i\right] = 0$, and $\beta_t = \beta$. While practitioners often assume the former, the latter indicates the time-invariant effect of covariates.

Proposition 3.1. *Under Assumptions 1–3 combined with Assumption 4 or 5, we can decompose μ_g as*

$$\mu_g = \sum_{\ell \in g}\left(\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] ATT_\ell\right) + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell \in g'}\left(\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] ATT_\ell\right) + \sum_{\ell \in g^{excl}}\left(\sum_{e=0}^{T} E\left[\omega_{i,e,\ell}^{g}\right] ATT_\ell\right)$$

Intuitively, TWFE regression (2.1) with a constant coefficient before $\mathbf{X}_{i,t}$ cannot capture the time-varying effect of covariates. In addition, (2.1) will mistake this time-varying effect for the treatment effect. This obscures the causal interpretation of μ_g even under treatment effect homogeneity.

The corresponding decomposition for fully dynamic specification (2.2) with $ATT_{\ell'} = 0$, $\ell' \in g^{excl}$, the decomposition simplifies to

$$\mu_\ell = ATT_\ell$$

We take the fully dynamic specification (2.2) as DGP, where $v_{i,t}$ is uncorrelated with covariates in all time periods. As expected, the decomposition indicates an unbiased estimation of treatment effects under zero treatment effects for the excluded periods. This setting implies treatment effect homogeneity and time-invariant effect of covariates, which are crucial for coefficients to be causally interpretable.

Remark 3.1. For the specification of canonical two-period DID

$$Y_{i,t} = \alpha_i + \lambda_t + \mu \cdot D_{i,t} + \beta^{\intercal}\mathbf{X}_{i,t} + v_{i,t}, t = 0, 1$$

Under Assumptions 1–3 combined with zero treatment effects for excluded relative periods, the decomposition simplifies to

$$\mu = ATT + E\left[\omega_{i,1}^{0} \cdot E\left[Y_{i,1} - Y_{i,0}|\mathbf{X}_i, G_i = 0\right]\right]$$

where $G_i := 1\{D_{i,1} = 1\}$ and $ATT = E\left[Y_{i,1} - Y_{i,1}^{\infty}|G_i = 1\right]$. As long as Assumption 4 or 5 can be satisfied further, the decomposition further simplifies to $\mu = ATT$. Therefore, this covariate effect bias will not be eliminated only by a two-period setting.

4. Conclusions

This paper highlights the pitfalls associated with using dynamic TWFE regressions with time-varying covariates to conduct causal inferences. We find that, under a conditional parallel trend, the coefficient of a given lead or lag can be contaminated by the effects from other periods and by the time-varying effects of covariates. While the former cross-lag contamination has already been suggested by Sun and Abraham (2021), the latter covariate effect bias is introduced in this paper. We show that covariate effect bias cannot be eliminated even under treatment effect homogeneity or only by a two-period setting. Moreover, it remains in the canonical two-period DID as well.

Acknowledgements

This research is supported by the National Natural Science Foundation of China (No. 71873080).

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.econlet.2022.110604.

References

Abadie, A., 2005. Semiparametric difference-in-differences estimators. Rev. Econom. Stud. 72 (1), 1–19.
Athey, S., Imbens, G., 2022. Design-based analysis in difference-in-differences settings. J. Econometrics 226 (1), 62–79.
Borusyak, K., Jaravel, X., Spiess, J., 2021. Revisiting event study designs: robust and efficient estimation. arXiv e-print 2108.12419v1.
Callaway, B., Sant'Anna, P., 2021. Difference-in-differences with multiple periods. J. Econometrics 225 (2), 200–230.
de Chaisemartin, C., D'Haultfoeuille, X., 2020. Two-way fixed effects estimators with heterogeneous treatment effects. Amer. Econ. Rev. 110 (9), 2964–2996.
Goodman-Bacon, A., 2021. Difference-in-differences with variation in treatment timing. J. Econometrics 225 (2), 254–277.
Sant'Anna, P., Zhao, J., 2020. Doubly robust difference-in-differences estimators. J. Econometrics 219 (1), 101–122.
Sun, L., Abraham, S., 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. J. Econometrics 225 (2), 175–199.